Title of the Article

Khaled Surname*[1], Second Author[1], and Third Author[2]

[1]Lab-STICC, UMR CNRS 6285, ENSTA, Brest, France
[2]Other Institution, City, Country

Abstract

Write a concise summary (150–250 words): context, problem, proposed method, main results, and significance.

Keywords: keyword one; keyword two; keyword three; keyword four

1 Introduction

Context and motivation. Related work can be cited like this [1]. State the contributions clearly:

- first contribution;
- second contribution;
- third contribution.

The rest of the paper is organized as follows: **??** presents the method, **??** the results, and **??** concludes.

2 Geometry and signal model

Every channel used in this paper is, to first order, a combination of two fixed functions of one normalised coordinate. this section sets the geometry, derives that coordinate and lists what the platform actually measures.

2.1 Survey Geometry

The vehicle moves at speed v = 1.0 m/s along straight legs that cross a straight cable at an incidence (Figure 1a). Two tri-axial fluxgates are mounted at the water line on the vehicle's lateral axis, UNO on the port side and DUO on the starboard side, separated by the baseline b = 1 m. For each sensor we use a local CPA frame: the origin is the point of its track directly above the cable, x is the horizontal normal to the cable, $\hat{y}$ points along the cable and $\hat{z}$ points down (Figure 1b). The cable then lies at x = 0, z = d, where d is the standoff.

*Corresponding author: email@ensta.fr

A sensor that crosses the cable at time t_c has horizontal distance $x = v(t - t_c)\sin$ from the cable at time t. Every field quantity below depends on x only through the ratio x/d, so we use the normalised coordinate

$$\tau = \frac{x}{d} = \frac{v\,(t - t_c)\sin\theta}{d} = \frac{t - t_c}{T_s}, \qquad T_s = \frac{d}{v\sin\theta}. \tag{1}$$

counts CPA distances, and T_s is the time the sensor takes to move one CPA distance across the cable; at $= 50°$ and v = 1 m/s it is 2.6 s for d = 2 m and 5.2 s for d = 4 m. T_s is the time scale of every matched template, and we call it the template scale. A leg that starts and ends at the lateral distance A from the cable covers |x| A, i.e. || K

with

$$K = \frac{L}{d}, \qquad L = A \text{ for a snake leg.} \tag{2}$$

L is the half-length of the pass measured across the cable. With the snake amplitude A = 6 m used in closed loop and d = 2–4 m, K = 3–1.5: a snake leg sees only a few CPA distances on each side of the crossing.

Acknowledgments

Funding sources and acknowledgments.

Data Availability

Statement on data and code availability.

References

[1] John Doe and Jane Smith. “An Example Article Title”. In: *Journal Name* 12.3 (2024), pp. 100–115. DOI: 10.0000/example.2024.001.

A Proof or Additional Derivations

Supplementary material.

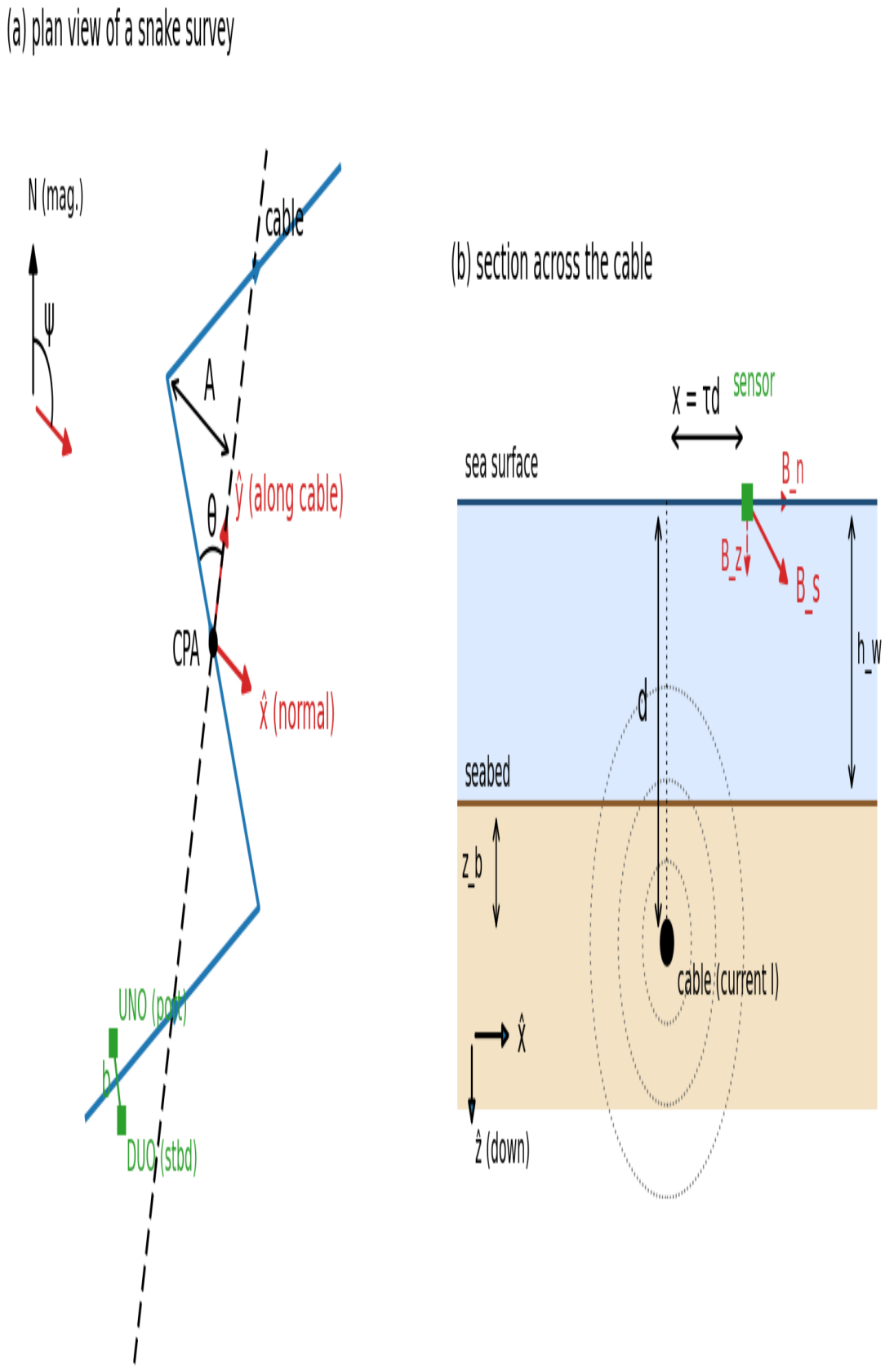


Figure 1: Survey geometry. (a) Plan view of a snake survey: cable(dashed), legs crossing it at incidence with amplitude A, CPA frame (x normal, ŷ along the cable), the two sensors UNO and DUO separated by the baseline b, and the angle from magnetic north to the cable normal. (b) Section across the cable: the sensor at the surface, the cable at standoff d below it (burial depth z_b below a seabed at water depth h_w), the normal coordinate x = d, and the azimuthal cable field B_s with its normal and vertical components